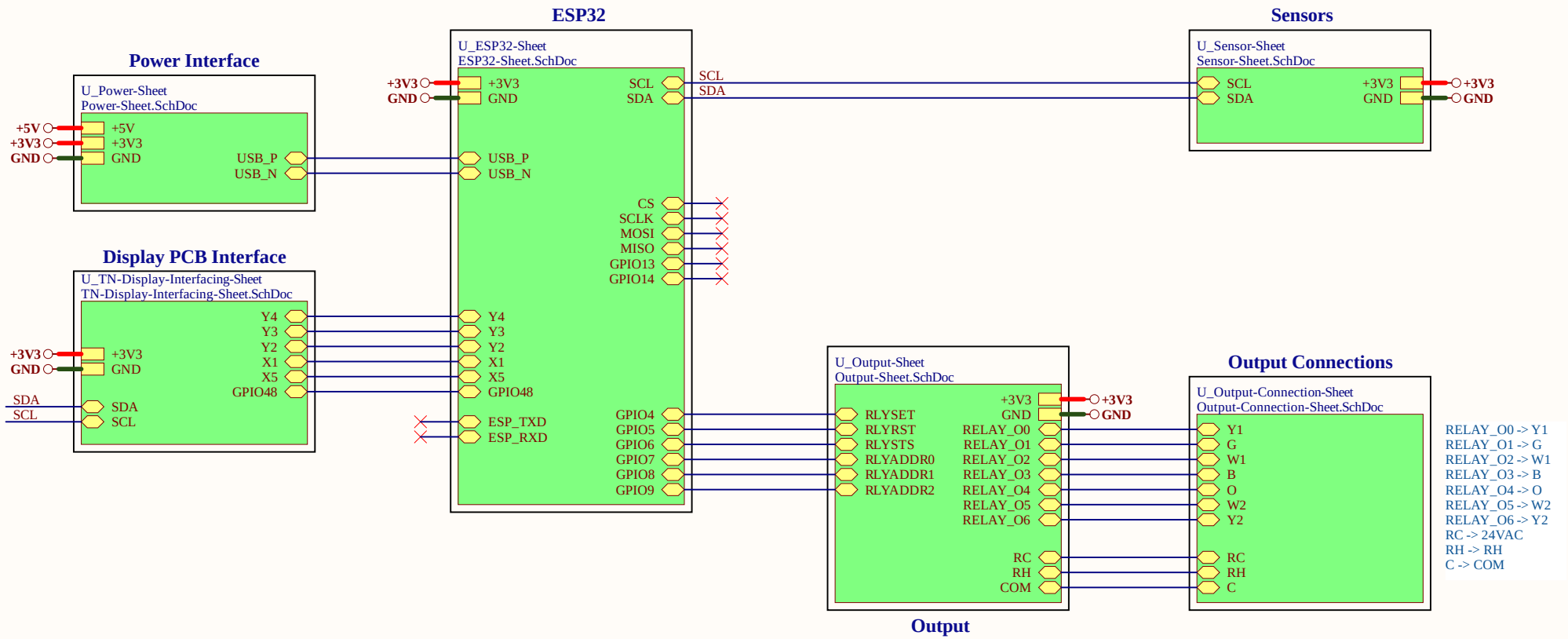


Relays, depending on the channel, switch between COM and 24VAC or RH wire.

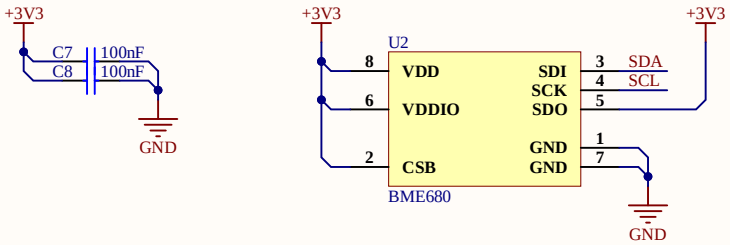
Y, G, O are switched betwen COM and 24VAC (RC)
W, B are switched between COM and RH wire

If C-wire is absent, 24VAC is absent, only W and B are switched to RH wire.

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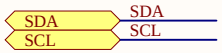
Title: Top-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 27/09/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 14

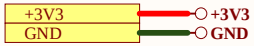
4-in-1 VOC Humidity Pressure Temperature



CSB -> VDDIO = I2C active
CSB -> GND = SPI active
I2C ADDR = BIN 1110111 HEX 0x77
I2C pull-up resistors are on the ESP32 sheet.

BSEC: Bosch Sensortec Environmental Cluster
BSEC precisely performs several calculations outside the device such as ambient air temperature, ambient relative humidity, pressure and air quality (IAQ) level.
https://www.bosch-sensortec.com/software-tools/software/bsec/
https://github.com/BoschSensortec/BSEC-Arduino-library

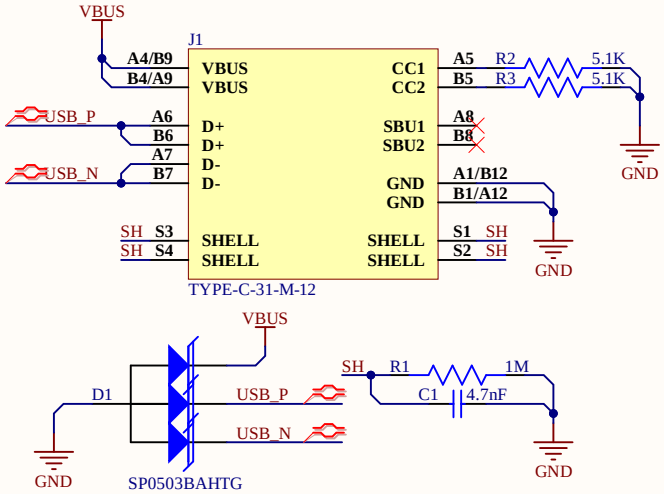




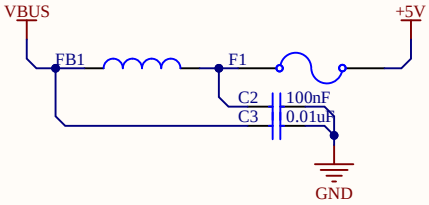
eSpark Consulting Group

Title: Sensor-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 27/09/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 14

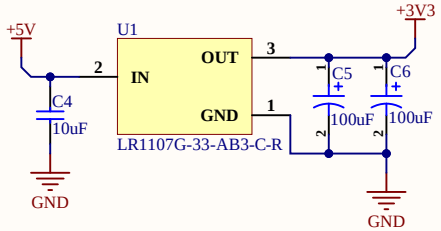
USB



Filters



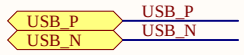
LDO 3.3V

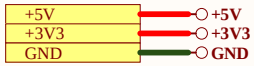


Est. Consumptions:
Idle: 160 mA
Max: 1050mA (worst case!)

Test Points





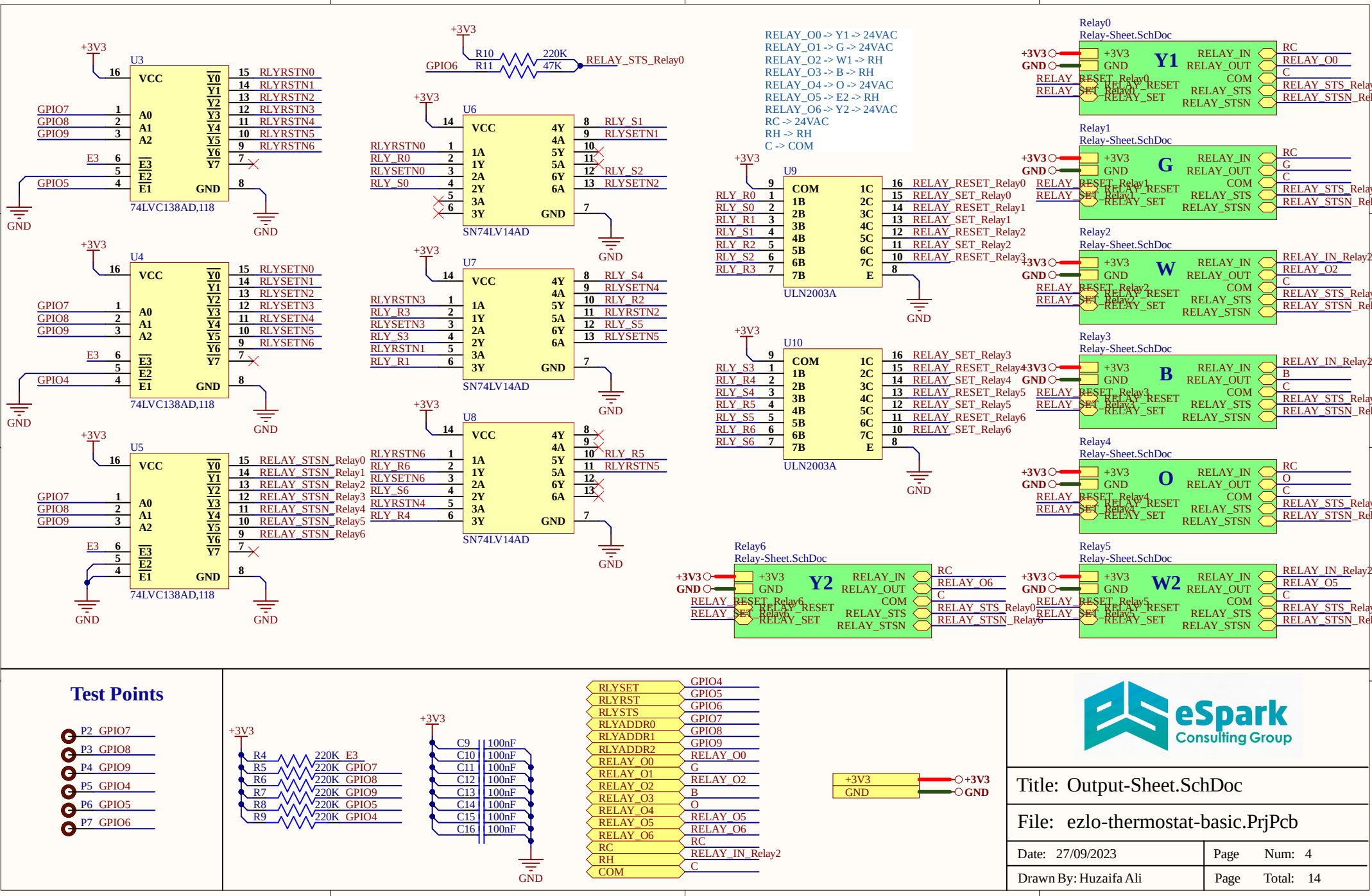


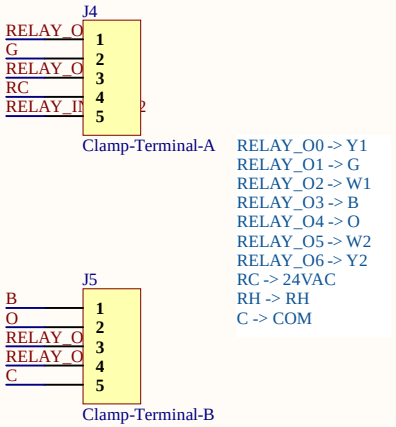
eSpark Consulting Group

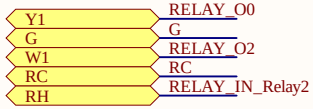
Title: Power-Sheet.SchDoc

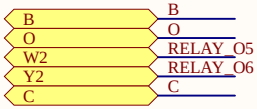
File: ezlo-thermostat-basic.PrjPcb

Date: 27/09/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 14


U3
74LVC138AD,118
U4
74LVC138AD,118
U5
74LVC138AD,118
U6
SN74LV14AD
U7
SN74LV14AD
U8
SN74LV14AD
U9
ULN2003A
U10
ULN2003A
RELAY_O0 -> Y1 -> 24VAC
RELAY_O1 -> G -> 24VAC
RELAY_O2 -> W1 -> RH
RELAY_O3 -> B -> RH
RELAY_O4 -> O -> 24VAC
RELAY_O5 -> E2 -> RH
RELAY_O6 -> Y2 -> 24VAC
RC -> 24VAC
RH -> RH
C -> COM
Relay0
Relay-Sheet.SchDoc
Y1
Relay1
Relay-Sheet.SchDoc
G
Relay2
Relay-Sheet.SchDoc
W
Relay3
Relay-Sheet.SchDoc
B
Relay4
Relay-Sheet.SchDoc
O
Relay6
Relay-Sheet.SchDoc
Y2
Relay5
Relay-Sheet.SchDoc
W2
Test Points
P2 GPIO7
P3 GPIO8
P4 GPIO9
P5 GPIO4
P6 GPIO5
P7 GPIO6
eSpark
Consulting Group
Title: Output-Sheet.SchDoc
File: ezlo-thermostat-basic.PrjPcb
Date: 27/09/2023
Page Num: 4
Drawn By: Huzaifa Ali
Page Total: 14

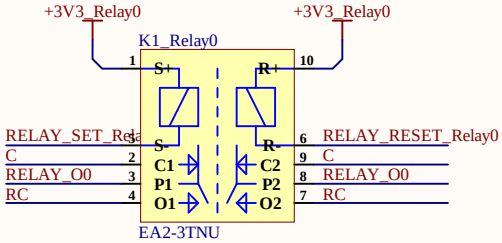


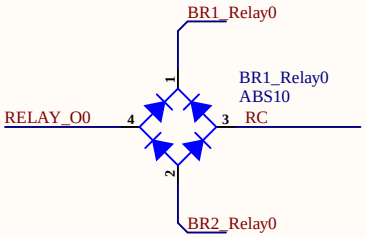


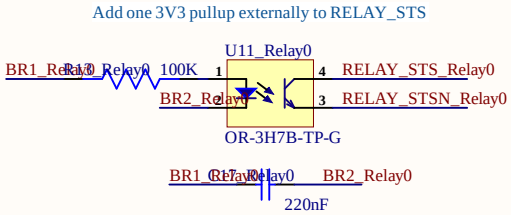


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Title: Output-Connection-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 27/09/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 14









Connect RELAY_IN to 24VAC or RH
AC Input





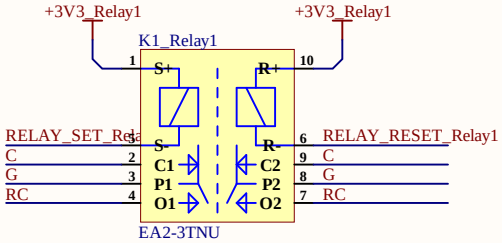


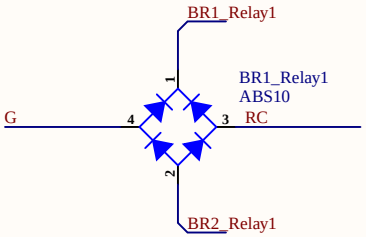
eSpark Consulting Group

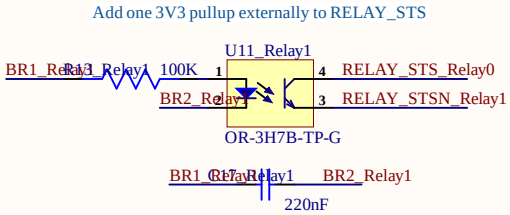
Title: Relay-Sheet.SchDoc

File: ezlo-thermostat-basic.PrjPcb

Date: 27/09/2023	Page Num: 6.1
Drawn By: Huzaifa Ali	Page Total: 14









Connect RELAY_IN to 24VAC or RH
AC Input





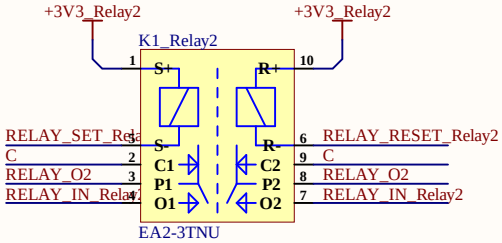


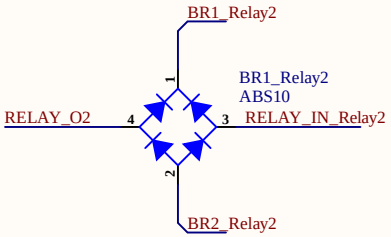
eSpark Consulting Group

Title: Relay-Sheet.SchDoc

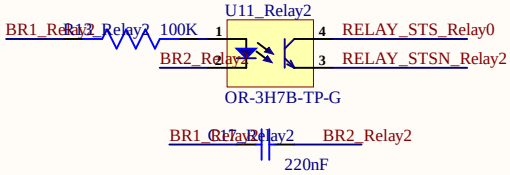
File: ezlo-thermostat-basic.PrjPcb

Date: 27/09/2023	Page Num: 6.2
Drawn By: Huzaifa Ali	Page Total: 14





Add one 3V3 pullup externally to RELAY_STS





Connect RELAY_IN to 24VAC or RH
AC Input





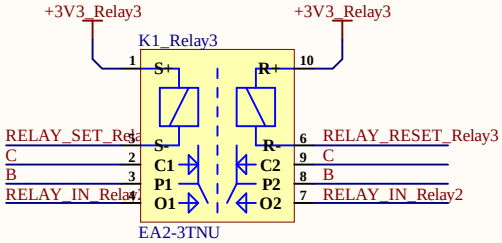


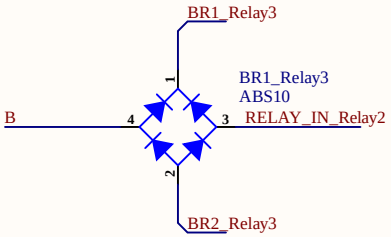
eSpark Consulting Group

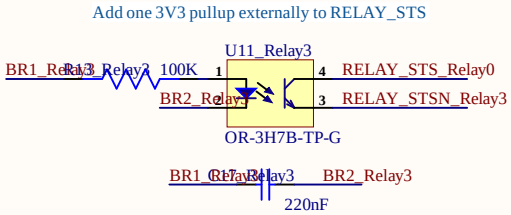
Title: Relay-Sheet.SchDoc

File: ezlo-thermostat-basic.PrjPcb

Date: 27/09/2023	Page Num: 6.3
Drawn By: Huzaifa Ali	Page Total: 14









Connect RELAY_IN to 24VAC or RH
AC Input





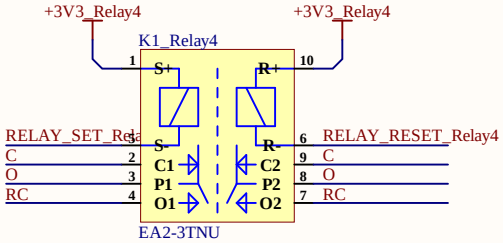


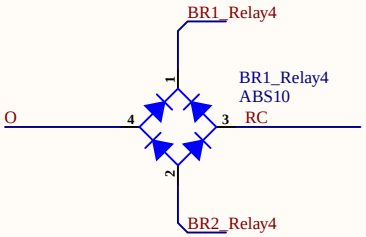
eSpark Consulting Group

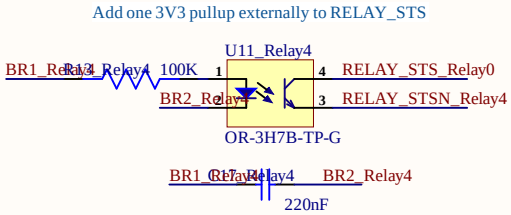
Title: Relay-Sheet.SchDoc

File: ezlo-thermostat-basic.PrjPcb

Date: 27/09/2023	Page Num: 6.4
Drawn By: Huzaifa Ali	Page Total: 14



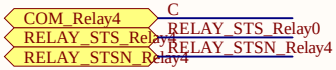






Connect RELAY_IN to 24VAC or RH
AC Input





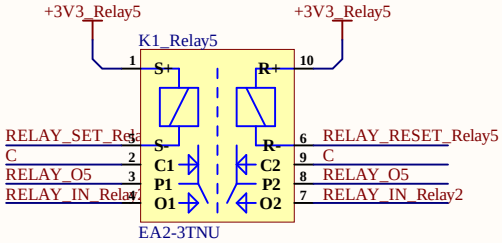


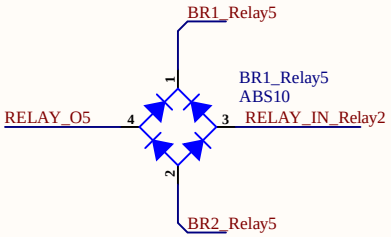
eSpark Consulting Group

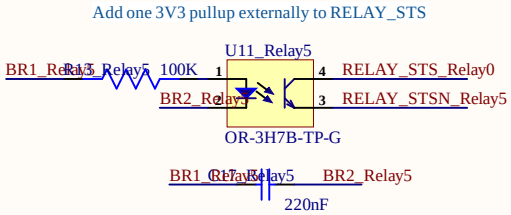
Title: Relay-Sheet.SchDoc

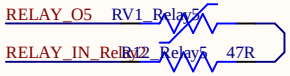
File: ezlo-thermostat-basic.PrjPcb

Date: 27/09/2023	Page Num: 6.5
Drawn By: Huzaifa Ali	Page Total: 14

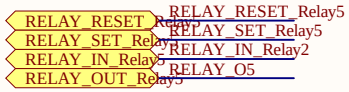








Connect RELAY_IN to 24VAC or RH
AC Input





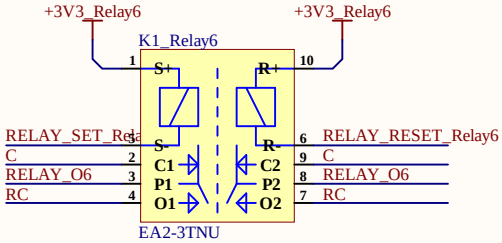


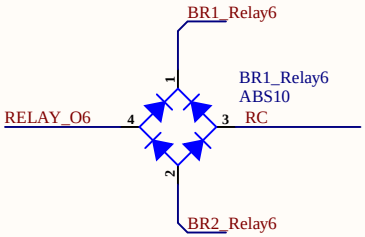
eSpark Consulting Group

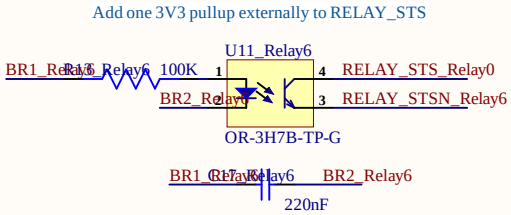
Title: Relay-Sheet.SchDoc

File: ezlo-thermostat-basic.PrjPcb

Date: 27/09/2023	Page Num: 6.6
Drawn By: Huzaifa Ali	Page Total: 14









Connect RELAY_IN to 24VAC or RH
AC Input







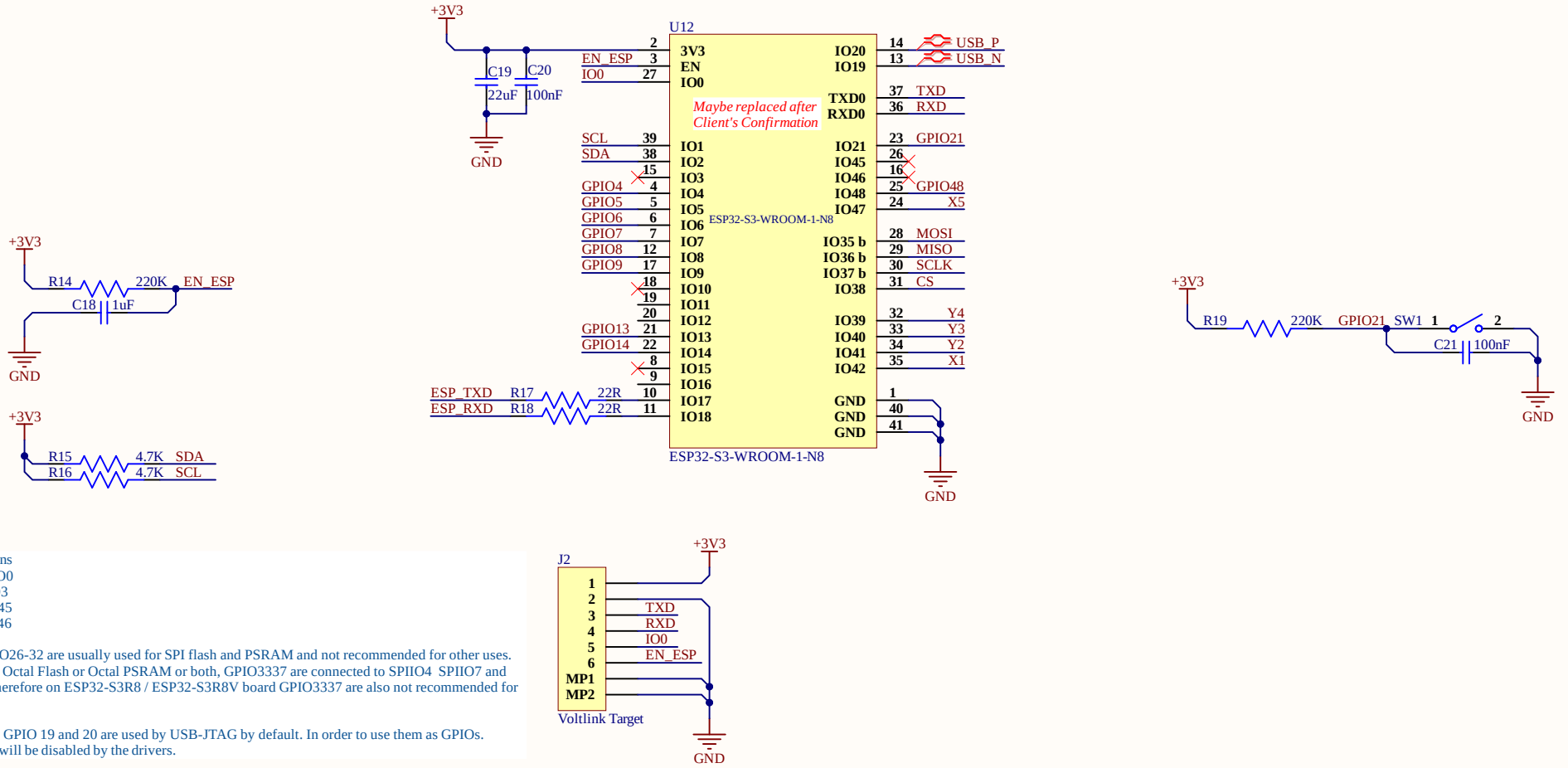
eSpark Consulting Group

Title: Relay-Sheet.SchDoc

File: ezlo-thermostat-basic.PrjPcb

Date: 27/09/2023	Page Num: 6.7
Drawn By: Huzaifa Ali	Page Total: 14

ESP32 MCU

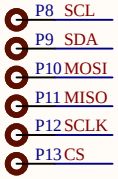


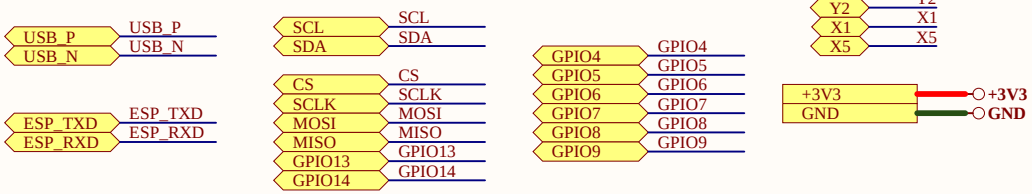
Strapping pins
- BOOT = IO0
- JTAG = IO3
- VSPI = IO45
- LOG = IO46

SPI0/1: GPIO26-32 are usually used for SPI flash and PSRAM and not recommended for other uses. When using Octal Flash or Octal PSRAM or both, GPIO3337 are connected to SPIIO4 SPIIO7 and SPIDQS. Therefore on ESP32-S3R8 / ESP32-S3R8V board GPIO3337 are also not recommended for other uses.

USB-JTAG: GPIO 19 and 20 are used by USB-JTAG by default. In order to use them as GPIOs. USB-JTAG will be disabled by the drivers.

Test Points



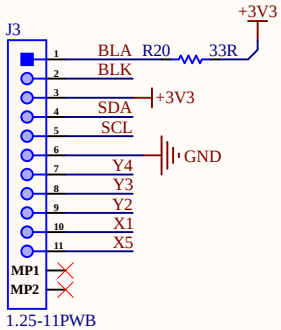


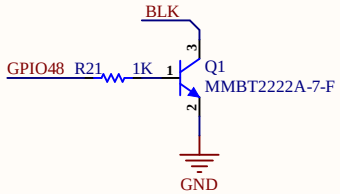
eSpark Consulting Group

Title: ESP32-Sheet.SchDoc

File: ezlo-thermostat-basic.PrjPcb

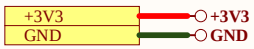
Date: 27/09/2023	Page Num: 8
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Title: TN-Display-Interfacing-Sheet.SchDoc	
File: ezlo-thermostat-basic.PrjPcb	
Date: 27/09/2023	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 14